



Derivatives Pricing in DeFi: Valuation of Automated Option Vaults

CONTENTS

1. Executive Summary 1

2. Mathematical Framework 1

 2.1 Black-Scholes Model 1

 2.2 The Greeks 1

3. Pricing Analysis 1

 3.1 Theoretical Valuation..... 1

 3.2 Risk Sensitivities (Visualized)..... 2

4. Conclusion 2

1. EXECUTIVE SUMMARY

As institutional capital enters DeFi, the demand for hedging instruments has surged. "DeFi Option Vaults" (DOVs) automate the selling of covered calls to generate yield. This project implements a **Black-Scholes-Merton** pricing engine to value ETH options and calculate risk sensitivities ("The Greeks").

For an Out-of-the-Money (OTM) ETH Call Option (Strike \$2,600, Expiring in 30 days), the model derived a fair value of **\$133.35**. Furthermore, the sensitivity analysis highlights the convexity of the position, with a Delta of **0.45**, indicating the specific hedge ratio required to neutralize directional risk.

2. MATHEMATICAL FRAMEWORK

2.1 BLACK-SCHOLES MODEL

The fair price of a European Call Option $C(S, t)$ is governed by:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

Where:

- S_0 : Current Asset Price (\$2,500)
- K : Strike Price (\$2,600)
- σ : Annualized Volatility (60%)
- r : Risk-free Rate (5%)
- T : Time to Maturity (30/365 years)

2.2 THE GREEKS

To manage the risk of an option portfolio, we calculate partial derivatives:

- **Delta** ($\Delta = \frac{\partial C}{\partial S}$): Sensitivity to spot price changes.
- **Gamma** ($\Gamma = \frac{\partial^2 C}{\partial S^2}$): Rate of change of Delta (Convexity).

3. PRICING ANALYSIS

3.1 THEORETICAL VALUATION

Based on the input parameters ($S = \$2500, K = \$2600, \sigma = 60\%$), the model outputs:

- **Call Option Price: \$133.35**
- **Yield Generation:** If a DOV sells this option weekly, it generates annualized yield for depositors, compensated by the risk of the option expiring In-the-Money (ITM).

3.2 RISK SENSITIVITIES (VISUALIZED)

Figure 1 illustrates the dynamic nature of option pricing.

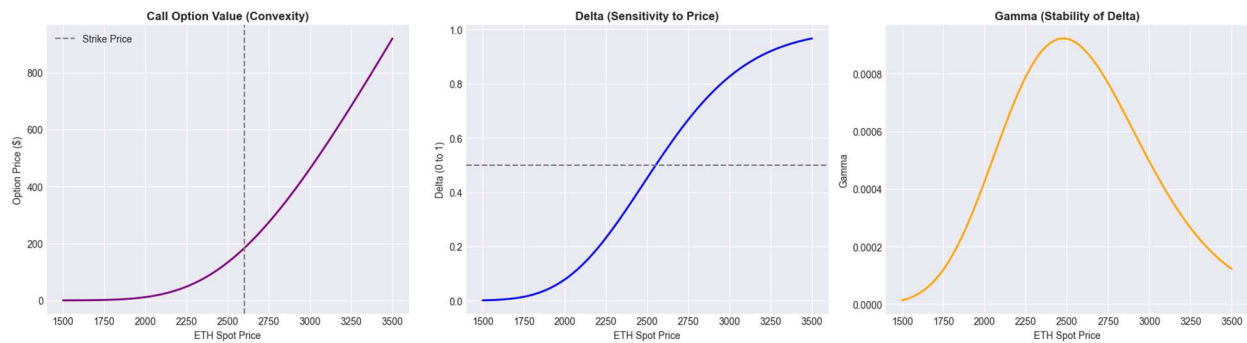


Figure 1: Sensitivity Analysis.

- **Left (Price):** Shows the non-linear (convex) payoff profile. The option gains value exponentially as ETH approaches the strike.
- **Center (Delta):** The Delta is **0.4530**. This means for a \$1 move in ETH, the option moves by \$0.45. To be "Delta Neutral," a market maker would need to short 0.453 ETH against this position.
- **Right (Gamma):** Gamma peaks exactly at the Strike Price (\$2,600). This is the "danger zone" for hedgers, where Delta changes most rapidly, requiring frequent rebalancing.

4. CONCLUSION

This project built a robust pricing engine for DeFi derivatives. The results validate that crypto options command high premiums primarily due to the elevated volatility parameter ($\sigma = 60\%$) compared to traditional equities. For DeFi protocols, this model serves as a foundational component for:

1. **Automated Market Making for Options (e.g., Lyra).**
2. **Impermanent Loss Hedging** (Using puts to offset LP downside).
3. **Structured Products** (Principal-protected yield vaults).